



YAQIDH: AI-BASED VIOLENCE DETECTION SYSTEM

for Schools, Care Institutions, and Public Spaces



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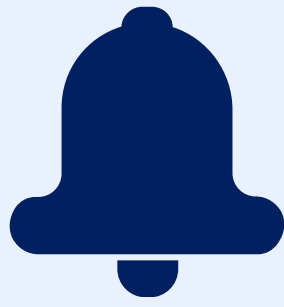
Introduction

Public safety is increasingly threatened by violent incidents in public areas such as schools and care centers, while traditional CCTV monitoring is slow, error-prone, and inefficient. This project proposes **a real-time AI-based violence detection system** that **analyzes** live video to detect aggressive behaviors and abnormal actions, **sends** instant alerts to authorities for rapid response, classifies incidents by severity (low, medium, high), **logs** all events with details such as time, location, camera ID, and severity, and provides an interactive dashboard for real-time monitoring.

WHY IT MATTERS



Saudi Arabia works hard to reduce violence in public spaces through emergency numbers (911/999), strong security presence, surveillance cameras, and strict laws against offenders.



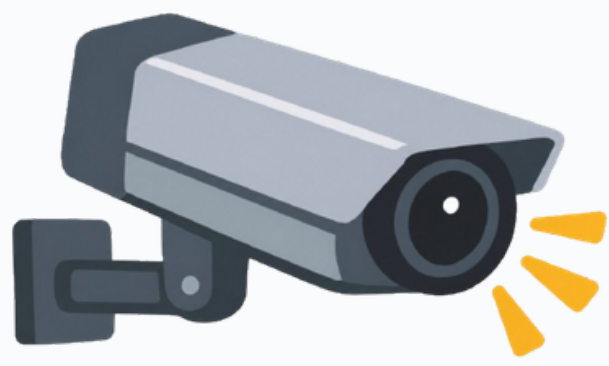
However, these measures can still be reactive and sometimes slow to respond in fast-moving situations.



But here's the thing Violence can happen in seconds, and every second of delay can make a critical difference.

Methodology

1. Input: Real-time Surveillance Video

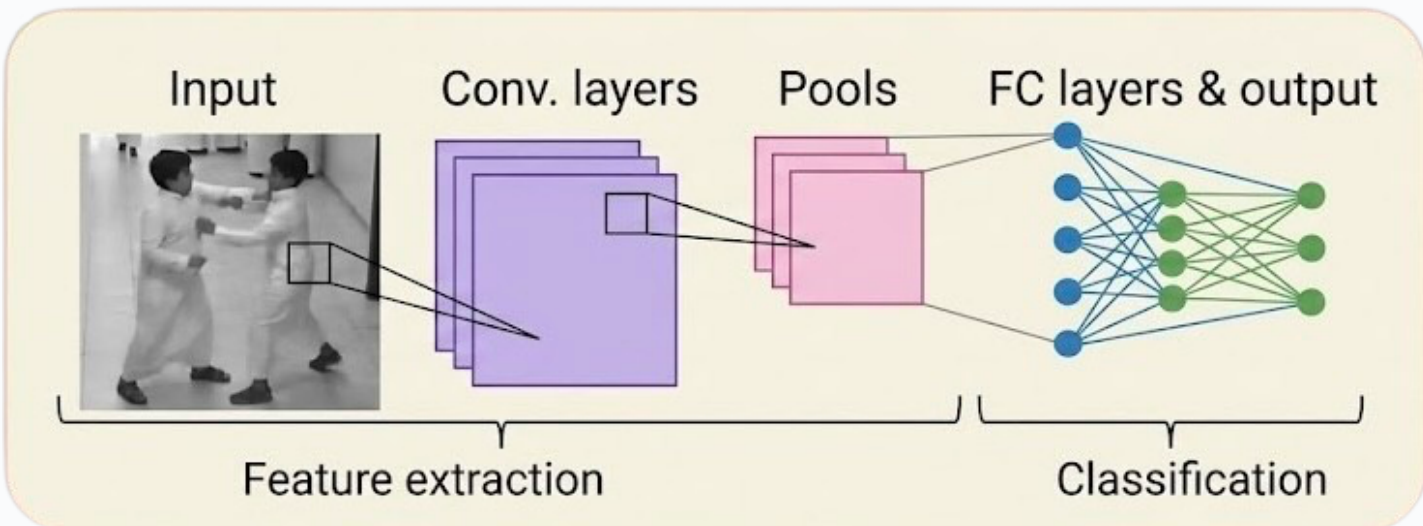


2. Dats Pre-processing

- Frame Extraction
- Noise Removal
- Frame Enhancement
- Pose Estimation



3. Deep learning models



4. Classification

- Violenc ? Yes/No
- Severity Level:

Low Medium High



Alter

5. violence Metadata Storage



- location
- Time/date
- Severity Level
- Camera IP

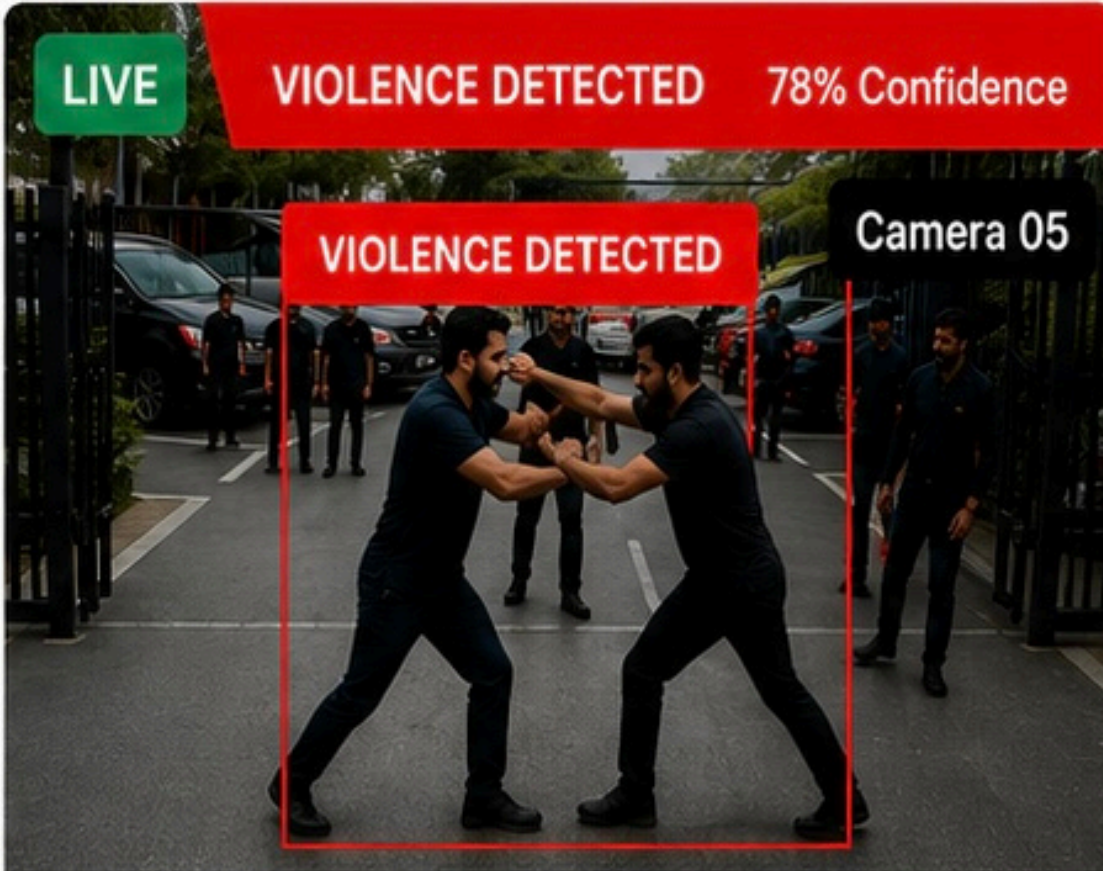
LIVE MONITORING SYSTEM

6 Cameras Active 2 Active Alerts

All Cameras 2x3 Grid



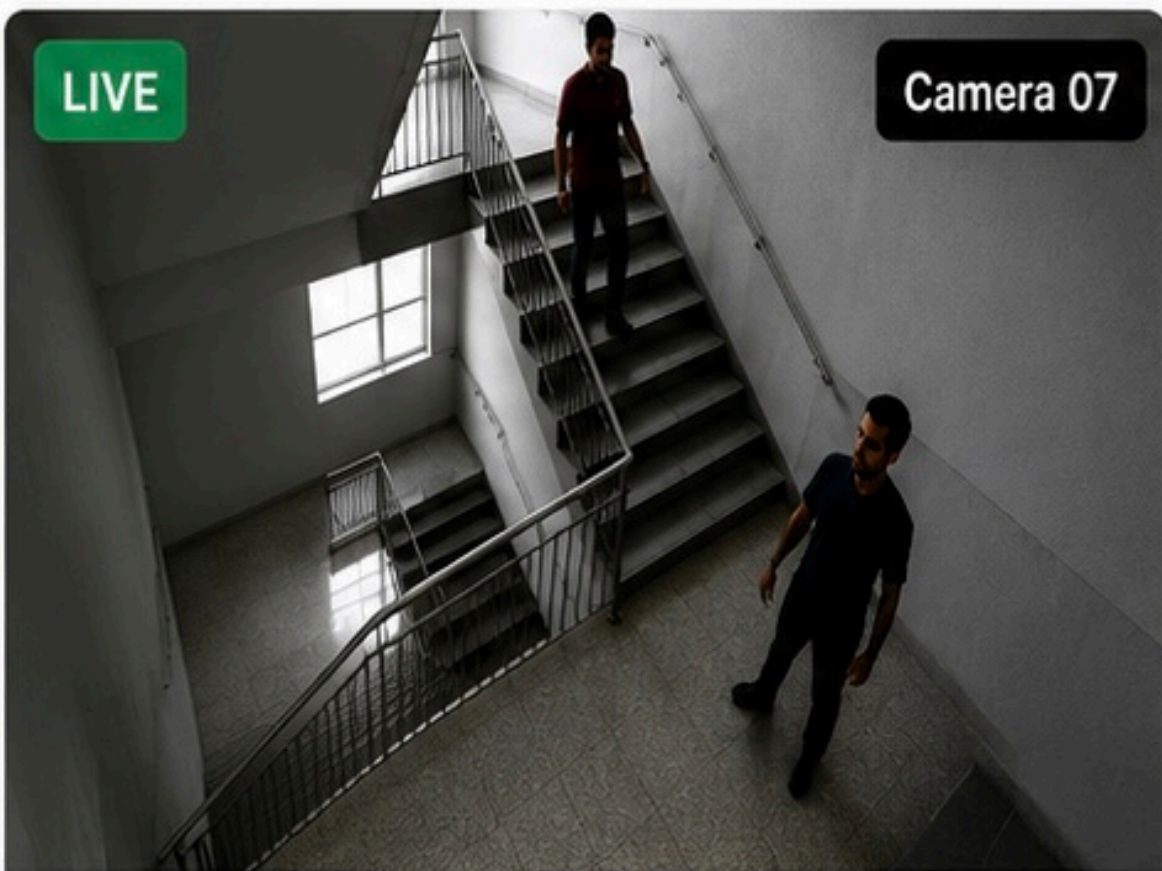
Cafeteria
Ground Floor



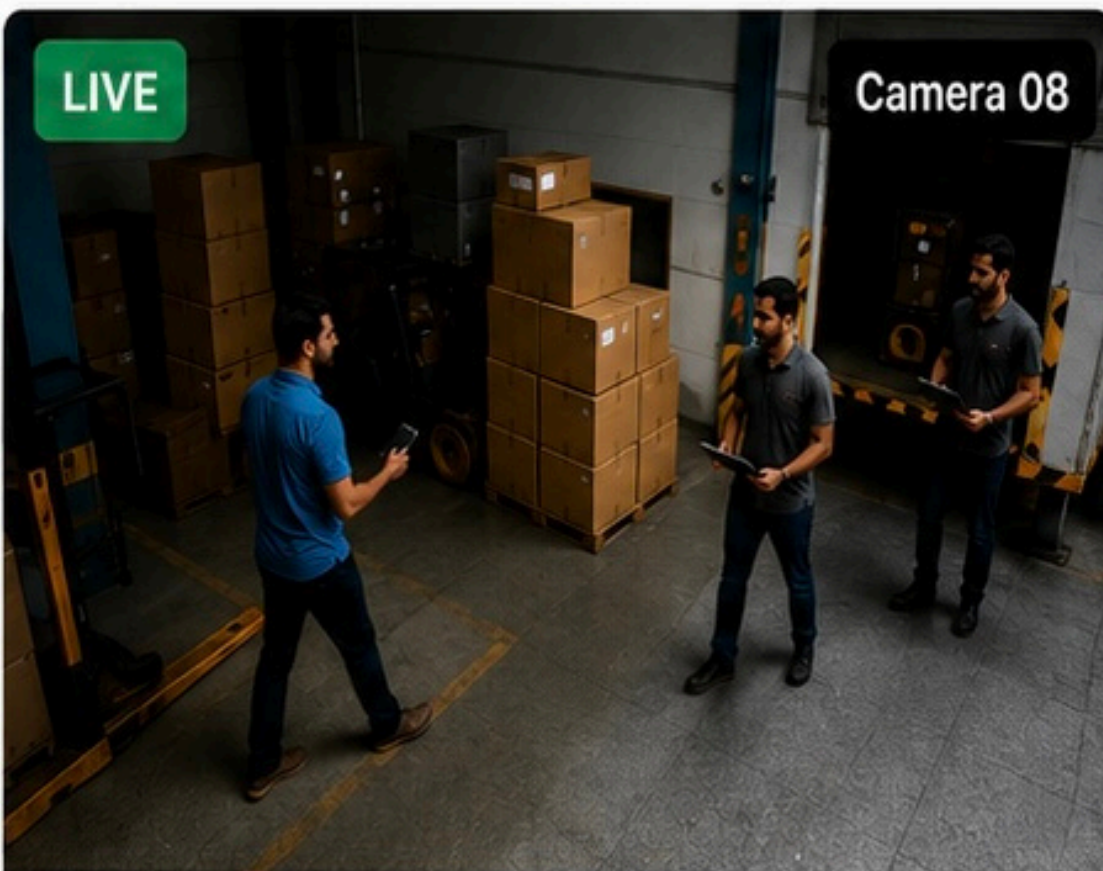
Parking Lot B
Parking Area



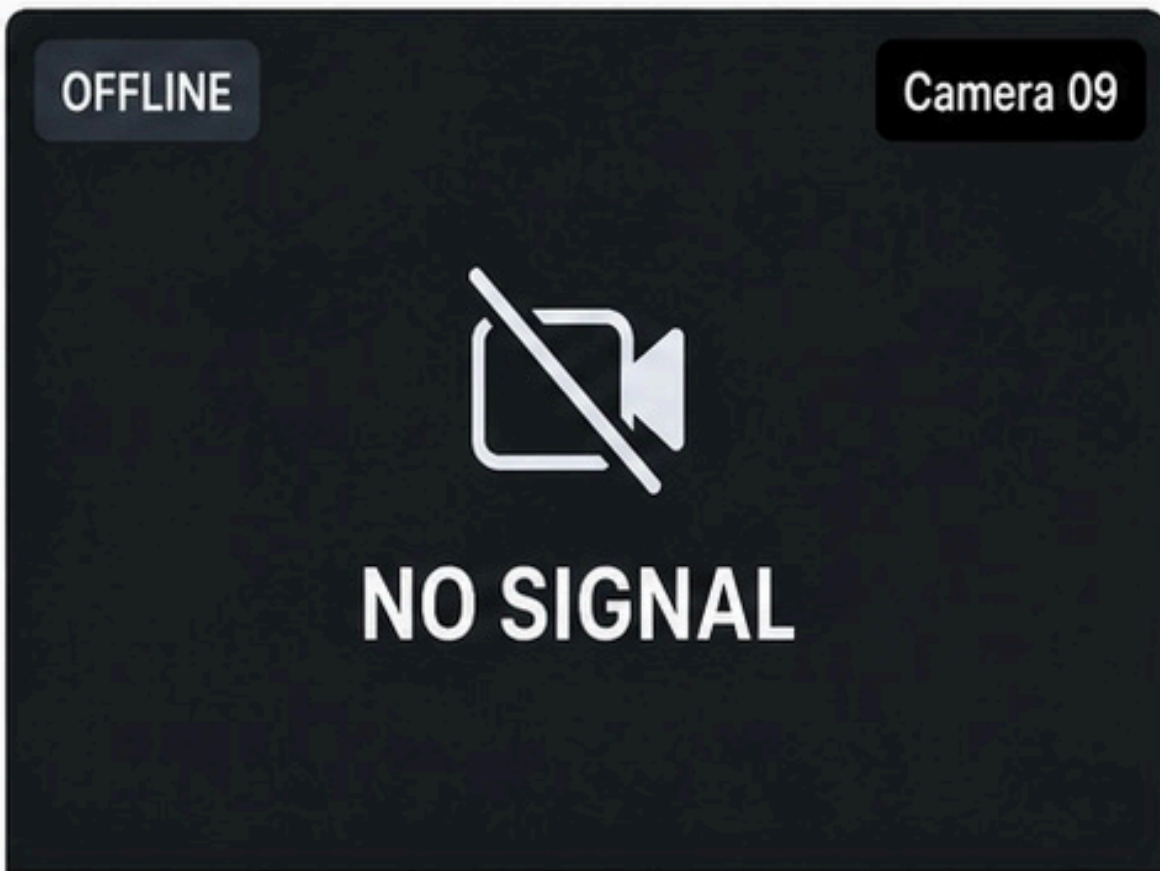
Reception
Main Lobby



Stairwell A
Floor 1-3

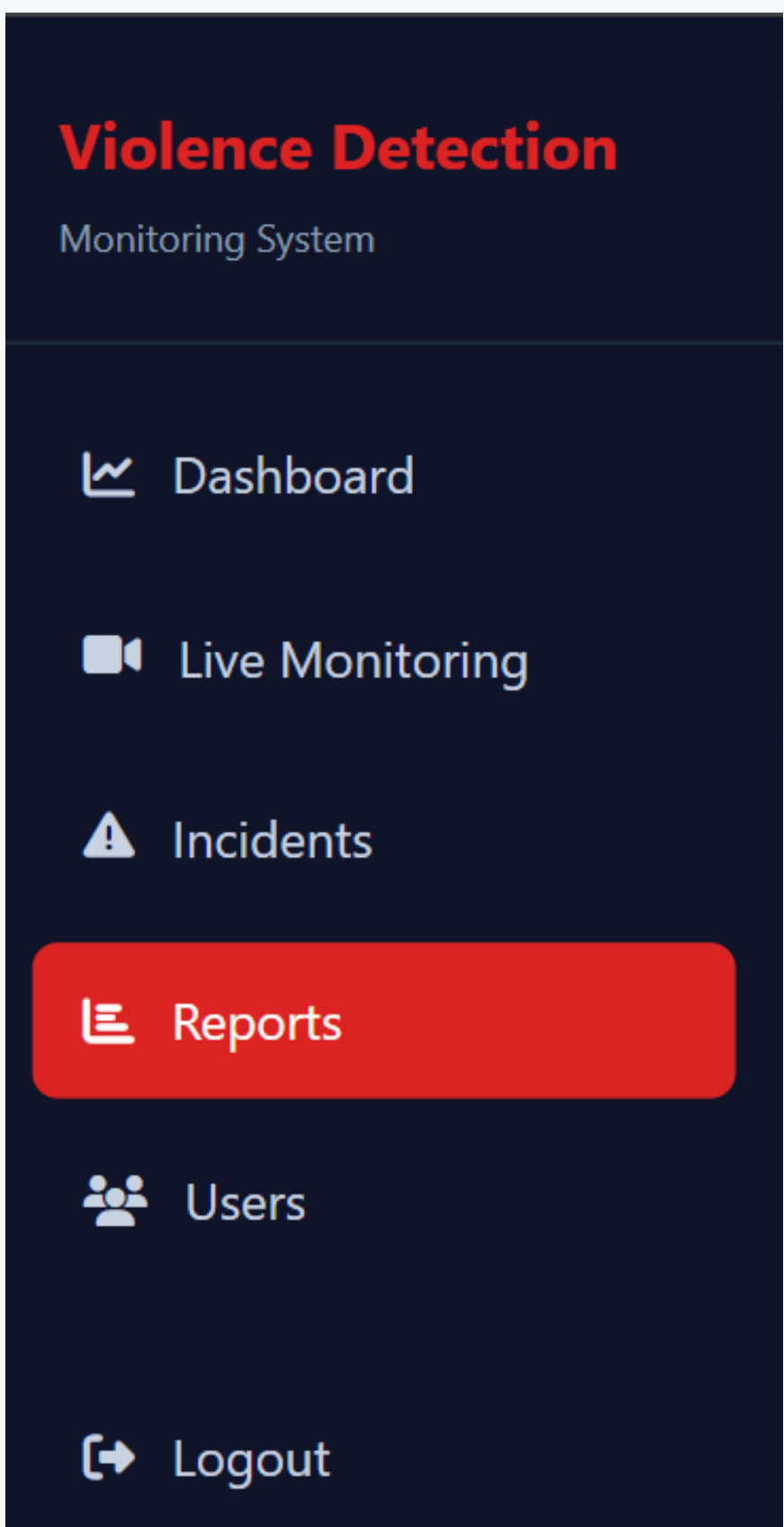


Loading Dock
Rear Entrance



Server Room
Basement

REPORTS



Reports & Analytics

Statistical analysis and incident trends

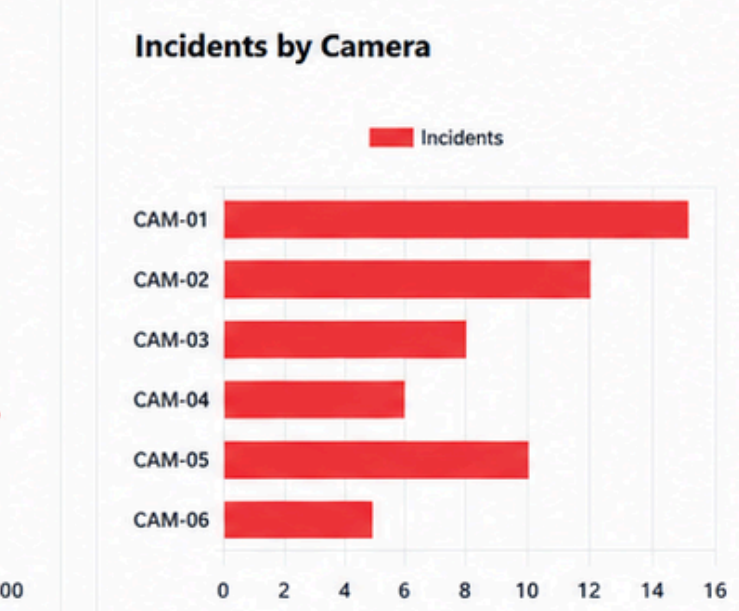
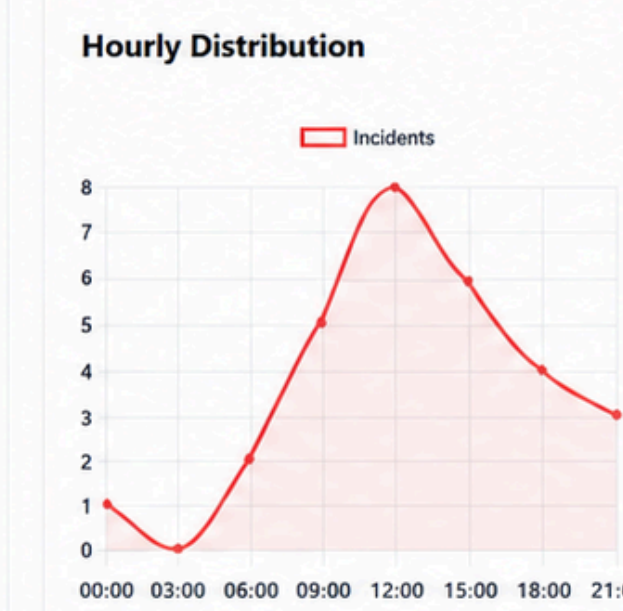
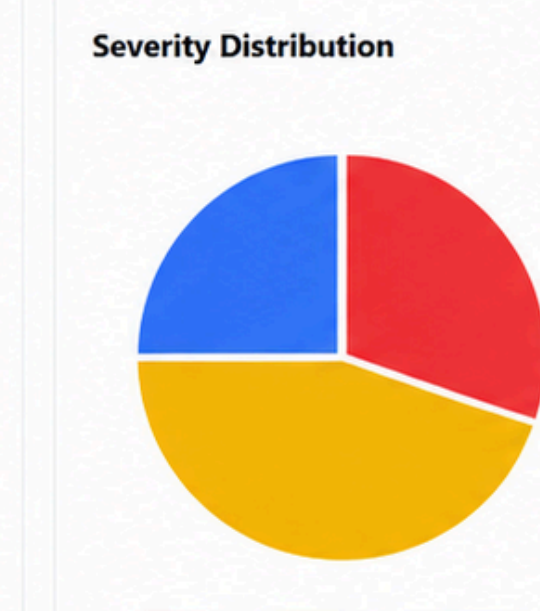
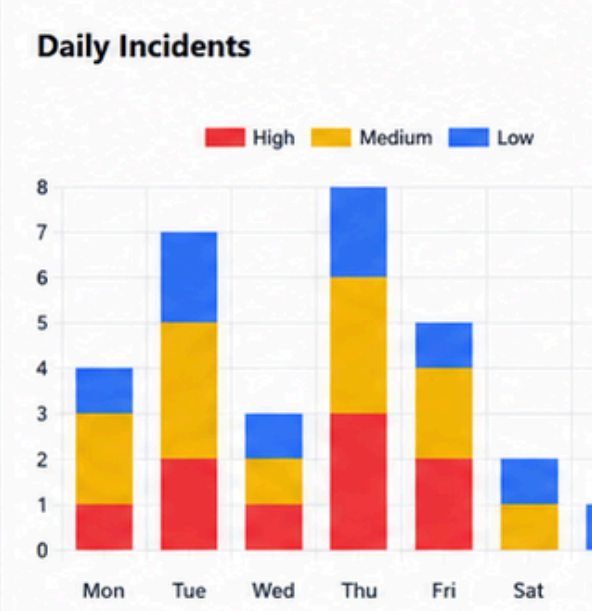
Last 7 Days Generate Report

Total Incidents
40
↓ 12% less than last week

High Severity
12
30% of total

Avg Response Time
3.2m
↑ 8% slower than average

Detection Accuracy
94.2%
Last 30 days



Peak Hour Alert
12:00 - 15:00
Highest incident frequency window

Top Location
Camera 01
Entrance Gate - 15 incidents

Improvement
-12%
Incident reduction vs last period

CONCLUSION

The system provides a scalable and intelligent solution for violence detection. It surpasses existing methods by leveraging advanced AI models to classify events, store incidents, and generate alerts in real time. The system supports Saudi Vision 2030 goals for building smart and secure environments.

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